

The Well Water Owner's Handbook

The Complete Guide to Testing, Understanding, and Treating Your Private Well Water

By the WellWater HQ Team (Trilot) 2026 Edition

How to Use This Book

You don't need to read this cover to cover before your next glass of water. Most people pick this up for one of three reasons: you just moved into a house with a private well and nobody explained what that means, you got a lab report back and don't know what the numbers mean, or you're noticing a taste, smell, or stain that's making you nervous.

Whatever brought you here, use the book that way. If you have a lab report in hand right now, skip to Chapter 4. If you're buying a house with a well next month, go straight to Chapter 9. If you just want to know what to test for and when, Chapter 2 is your calendar.

But if you have an evening to spare, read it in order once. Well water is not complicated once you understand a handful of ideas, and every chapter after Chapter 1 leans on that foundation. The well owners who avoid health scares and expensive treatment mistakes aren't the ones who read the most about hydrogeology. They're the ones who test on a schedule, understand what their results actually mean, and treat the specific problem their water has instead of buying whatever system a salesperson is pushing.

One more thing. Nothing in this book replaces an actual water test from a certified laboratory, and nothing in it is medical advice. Private well water is entirely your responsibility; no utility is testing it for you. Local geology, agricultural activity, and regulations vary enormously by county and state. Treat this as the framework that helps you ask the right questions, order the right tests, and recognize a bad answer when you hear one.

Table of Contents

1. How Your Well and Water Supply Actually Work
 2. The Testing Calendar
 3. Reading a Water Test Report
 4. Common Contaminants and What They Mean
 5. Choosing a Certified Lab and Collecting a Sample Correctly
 6. Treatment Systems: Matching the Fix to the Problem
 7. Troubleshooting Taste, Smell, Color, and Pressure Problems
 8. Seasonal Care and Flood/Emergency Protocols
 9. Buying or Selling a Home with a Private Well
 10. Costs & Budgeting
 11. Records, DIY vs. Pro, and Building a Relationship with a Water Professional Appendix A: Printable Testing Log Appendix B: Glossary of Terms Appendix C: 20 Questions to Ask Before You Hire a Water Treatment Company Closing Note
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Chapter 1: How Your Well and Water Supply Actually Work

A private well is your own personal water utility, drilled into the ground and drawing from an aquifer beneath your property. There's no municipal treatment plant testing it, adding disinfectant, or monitoring it for contaminants day to day. Whatever is in the ground beneath your property, or upstream of it in the aquifer, is what comes out of your tap unless you test for it and treat it yourself. Understanding that single fact changes how you think about your water, because there is no safety net catching problems before they reach your glass. You are the water utility, whether you signed up for the job or not.

The well itself

Most modern private wells are drilled wells: a narrow borehole, typically 6 to 8 inches in diameter, drilled anywhere from 100 to 500+ feet deep depending on local geology, lined with a steel or PVC casing to keep the borehole from collapsing and to seal out shallow surface contamination. A submersible pump sits down in the well, pushing water up through a pipe to a pressure tank in your house, which maintains water pressure so the pump doesn't have to run every time you open a tap.

Older properties sometimes have dug wells (wide, shallow, hand-dug, generally more vulnerable to surface contamination) or driven point wells (shallow, narrow, driven into sand or gravel aquifers). Knowing which type you have matters, because shallower wells are generally more exposed to surface-level contamination like bacteria and agricultural runoff, and often warrant more frequent testing.

The well cap and casing

At the top of the well, a sanitary well cap seals the casing against insects, rodents, and surface water intrusion. This is a small, unglamorous part that matters enormously: a cracked, missing, or poorly sealed cap is one of the most common and easiest-to-fix sources of bacterial contamination in an otherwise sound well. The casing itself should extend at least a foot above grade and be sloped away from the well so surface water doesn't pool around it and seep down alongside the casing.

The aquifer

Water doesn't sit in an open underground lake. It moves through saturated layers of soil, sand, gravel, or fractured rock called an aquifer. Aquifer type shapes both water quality and vulnerability. A well drawing from a deep, confined aquifer (protected above by a layer of impermeable clay or rock) is generally more shielded from surface contamination than a shallow, unconfined aquifer where rainwater and whatever it picks up on the way down reaches the water table directly. Wells near agricultural land, septic systems, gas stations, or industrial sites carry contamination risks specific to what's happening upstream in the same aquifer, which is part of why "my neighbor's well is fine" tells you very little about your own.

What's naturally in groundwater

Groundwater picks up minerals and elements from the rock and soil it passes through long before any human activity gets involved. Hardness (calcium and magnesium), iron, manganese, and in some regions naturally occurring arsenic or radon, are geological facts of your local aquifer, not necessarily signs of contamination. This is why interpreting a water test correctly matters so much: a high iron reading might mean nothing more than "iron-rich bedrock in your area," while a coliform bacteria detection almost always means something entered the well that shouldn't have.

Why the Safe Drinking Water Act doesn't cover you

The EPA regulates public water systems under the Safe Drinking Water Act, setting enforceable limits and requiring regular testing and treatment. Private wells serving fewer than 25 people are explicitly excluded from this regulation. No agency is required to test your well, treat your water, or notify you of a problem. The EPA and CDC both recommend private well owners test their own water regularly, but "recommend" is the operative word. There is no compliance mechanism, no automatic notification, and no one coming to check unless you arrange it yourself.

A quick mental model

Picture your well as a straw stuck into a shared underground reservoir. Everything happening on the surface for miles around, farming, septic systems, old fuel tanks, road salt, industrial activity, eventually has the potential to reach that reservoir and, through it, your straw. Testing is how you find out whether anything has. Treatment is how you fix what testing finds. Every chapter in this book builds on that two-step loop: test regularly, treat what's actually there.

The bottom line: A private well draws directly from a local aquifer with no municipal treatment or oversight standing between the ground and your tap. Your well cap, casing, and surrounding drainage protect against surface contamination, but only regular testing tells you what's actually in your water, since it is entirely your responsibility to know and manage.

Chapter 2: The Testing Calendar

Well water testing isn't complicated, but like septic maintenance, it only works if you do it on a schedule instead of waiting for a taste, smell, or health scare to force your hand.

The baseline: annual testing

The EPA and CDC both recommend testing private wells at minimum once a year for total coliform bacteria and nitrates. This is the non-negotiable baseline for every private well owner, regardless of how good your water tastes or how long you've lived there. Bacteria and nitrate contamination are often invisible, tasteless, and odorless, which is exactly why annual testing matters even when nothing seems wrong.

Testing beyond the baseline

Test	Recommended frequency	Why
Total coliform bacteria	Annually	Indicator of surface or sewage contamination; can be invisible and odorless
Nitrate	Annually	Indicates agricultural runoff or septic contamination; critical for infants and pregnant women
pH, hardness, iron, manganese	Every 1-2 years, or once for baseline then as needed	Affects taste, staining, and plumbing/appliance wear; usually not a health risk
Total dissolved solids (TDS)	Every 1-2 years	General water quality indicator
Arsenic	Every 3-5 years, or based on regional risk	Naturally occurring in some bedrock; no taste or smell
Radon	Once for baseline, retest every 5 years or after well work	Naturally occurring gas in some regions; health risk from both water and air exposure
PFAS ("forever chemicals")	Once for baseline in at-risk areas, otherwise as regional guidance suggests	Emerging contaminant near industrial sites, airports, or firefighting foam use areas
Lead and copper	Once for baseline, especially if the home has older plumbing	Comes from plumbing/fixtures, not usually the aquifer itself
Full mineral/metals panel	Every 3-5 years, or when buying/selling	Broad snapshot of water chemistry

Test immediately, outside the regular schedule, if:

- You notice a sudden change in taste, odor, color, or clarity

- Anyone in the household experiences unexplained gastrointestinal illness
- You're pregnant or have an infant in the home and haven't tested nitrate recently
- Flooding has occurred near the well, or the well was submerged
- You've had any work done on the well (new pump, casing repair, new construction nearby)
- A nearby land use has changed (new agricultural operation, gas station, industrial facility, known contamination case in your area)
- You're buying or selling a home with a private well (Chapter 9)
- It's been more than a year since your last bacteria and nitrate test, full stop

Building your own testing calendar

A reasonable working calendar for a typical private well owner:

- **Every year:** Coliform bacteria and nitrate, at minimum. Many labs bundle this as a basic "well water package" for a modest cost.
- **Every 2-3 years:** Add a broader panel covering hardness, iron, manganese, pH, and TDS, especially if you notice any aesthetic issues (staining, taste, scale buildup).
- **Once, for baseline, then every 3-5 years or per regional guidance:** Arsenic, radon, and PFAS, since these are typically geological or industrial in origin and don't fluctuate as quickly as bacteria can.
- **Any time something changes:** See the immediate-test list above.

What to log

Keep a simple log of every test, ordered the same way each year if possible so results are genuinely comparable:

- Date of sample collection and which lab performed the test
- Every result, with the units and the EPA Maximum Contaminant Level (MCL) or health advisory level noted alongside it
- Any treatment system changes (new filter, new UV system, new water softener) and their install/maintenance dates
- Any unusual events: taste or odor changes, illness in the household, nearby land-use changes, flooding

Appendix A has a ready-to-use log template. Keeping it updated is the single most useful thing you can do to catch a slow-building problem before it becomes a health issue, and it's the document that will matter most if you ever sell the property.

The bottom line: Test for coliform bacteria and nitrate every year without exception, layer in a broader mineral and metals panel every few years, and test immediately any time your water changes or a life event (new baby, pregnancy, home sale, flooding) raises the stakes. A consistent log turns scattered test results into a pattern you can actually trust.

Chapter 3: Reading a Water Test Report

A lab report full of chemical names and numbers is intimidating the first time you see one. It doesn't need to be. Almost every report follows the same basic structure, and once you know what to look for, you can read your own results in a few minutes.

The three columns that matter

Most lab reports show, for each parameter tested: your result, the unit of measurement (mg/L, or milligrams per liter, is the most common; it's roughly equivalent to parts per million), and a reference level, usually the EPA Maximum Contaminant Level (MCL) for public water systems, listed for comparison even though it isn't legally enforceable on your private well.

A result below the MCL doesn't automatically mean "perfect," and a result above it doesn't automatically mean "emergency." MCLs are legal limits designed with a margin of safety for public systems serving large populations continuously. Use them as a serious reference point, not a pass/fail grade, and read Chapter 4 for what specific results actually mean for your situation.

Bacteria results: the one area that's black and white

Total coliform bacteria results are typically reported as "present" or "absent," sometimes with a colony count. Unlike most chemical parameters, this one has no acceptable gray zone for drinking water: any detection of total coliform, and especially any detection of *E. coli* specifically, means the water should not be consumed until the source is identified and resolved, and the well is disinfected and retested clean. Chapter 4 and Chapter 7 cover what to do if you get a positive result.

Understanding "non-detect"

Many parameters will show as "ND" or "non-detect," meaning the lab's testing method didn't find any of that substance above its detection limit. This is good news, but it's worth knowing that "non-detect" means "below what this specific test can measure," not necessarily "zero." Different labs and different test methods have different detection limits, which is one more reason to use a certified, reputable lab (Chapter 5).

A sample reading walkthrough

Say your report shows:

Parameter	Your result	Unit	EPA MCL / advisory
Total coliform	Absent	—	Must be absent
Nitrate	3.2	mg/L	10 mg/L
Iron	0.8	mg/L	0.3 mg/L (aesthetic, not health-based)

Parameter	Your result	Unit	EPA MCL / advisory
Hardness	180	mg/L	No MCL (aesthetic)
Arsenic	ND	mg/L	0.010 mg/L

Reading this: no bacteria present, good. Nitrate is comfortably under the health-based limit. Iron is above the EPA's aesthetic guideline (not a health standard, but staining and taste can result), which explains any rust-colored staining you may have noticed and points toward an iron filtration solution (Chapter 6) rather than a health concern. Hardness has no formal limit because it's not a health issue, just a nuisance and appliance-wear issue. Arsenic wasn't detected, which is reassuring given some regions' naturally elevated arsenic levels.

Health-based vs. aesthetic (secondary) standards

The EPA sets two categories of standards, and knowing the difference prevents a lot of unnecessary alarm or, worse, unnecessary complacency:

- **Primary (health-based) standards:** legally enforceable limits for public systems, covering things like bacteria, nitrate, arsenic, and lead, because exceeding them carries a documented health risk.
- **Secondary (aesthetic) standards:** non-enforceable guidelines for things like iron, manganese, hardness, and taste/odor/color, because exceeding them affects how water looks, tastes, or interacts with plumbing, but isn't a documented health hazard at typical levels.

A report full of "exceeds secondary standard" flags is an annoyance-and-plumbing conversation. A report with a primary standard exceedance, especially bacteria, nitrate, arsenic, or lead, is a health conversation, and Chapter 4 walks through what each one actually means.

The bottom line: Every lab report follows the same basic pattern: your result, the unit, and a reference level. Learn to distinguish health-based (primary) standards from aesthetic (secondary) ones, and treat any bacteria detection as an immediate, non-negotiable action item rather than a number to interpret.

Chapter 4: Common Contaminants and What They Mean

This chapter walks through the contaminants most private well owners actually encounter, what they mean, and what typically follows a detection. This is educational information, not a diagnosis of your specific water; always work from your own lab results and, for anything health-related, your own healthcare provider or state health department.

Total coliform bacteria and E. coli

Coliform bacteria are naturally present in soil and vegetation and aren't necessarily harmful themselves, but their presence in well water indicates a pathway exists for surface contamination to reach your water, which means more harmful organisms could follow the same path. E. coli specifically indicates fecal contamination and is treated as a more urgent finding. A positive result means: don't drink the water (boil it first, or use bottled water) until the source is found and the well is shock-chlorinated and retested clean. Common sources include a damaged well cap, a shallow well near a septic system or livestock area, or surface water intrusion after flooding.

Nitrate

Nitrate comes primarily from agricultural fertilizer runoff, animal waste, and septic system effluent. The EPA MCL is 10 mg/L, set specifically because of the risk to infants: nitrate above this level can cause methemoglobinemia ("blue baby syndrome") in infants under six months, a serious and potentially fatal condition. Households with infants, or anyone preparing infant formula with well water, should never use water testing above the nitrate MCL without switching to a tested alternative source. Adults generally tolerate somewhat higher levels without acute effects, but persistent elevated nitrate is a sign of contamination worth addressing at the source where possible (nearby septic system, agricultural runoff) in addition to treatment.

Arsenic

Naturally occurring in bedrock in certain regions (parts of New England, the Southwest, the upper Midwest, and elsewhere), arsenic has no taste, odor, or color, which is exactly why it requires a specific lab test to detect. The EPA MCL is 0.010 mg/L (10 parts per billion). Long-term exposure to elevated arsenic is associated with increased cancer risk and other health effects, which is why it's treated as a serious finding requiring treatment (typically reverse osmosis or specific adsorptive media) rather than a wait-and-see issue.

Lead and copper

Unlike most contaminants on this list, lead and copper in well water usually come from your own plumbing, not the aquifer: older lead pipes, lead solder on copper pipe joints (common before 1986), or brass fixtures with lead components. Corrosive water (low pH, low mineral content) leaches these metals from plumbing more readily. A positive result points toward plumbing remediation and/or corrosion-control treatment as much as source-water treatment, and testing should include a "first draw" sample (water that's been sitting in the pipes overnight) to capture the worst-case exposure.

Radon

A naturally occurring radioactive gas from decaying uranium in soil and rock, radon in well water is a concern both because you can ingest it and because it off-gasses into indoor air, particularly during showering, where inhalation is the primary health concern. There's no federal MCL specifically for radon in water as of this writing, but health advisories exist, and regions with granite bedrock or known elevated radon (check your state's radon map) warrant baseline testing. Treatment typically involves aeration systems that off-gas the radon before it reaches your taps.

PFAS ("forever chemicals")

Per- and polyfluoroalkyl substances are a family of synthetic chemicals used in firefighting foam, non-stick coatings, and various industrial applications. They're called "forever chemicals" because they don't break down naturally and accumulate in the body over time. Wells near airports, military bases, firefighting training facilities, or certain industrial sites carry elevated risk. The EPA has been tightening enforceable limits for public water systems in recent years; check current guidance for your state, since this is an evolving regulatory area. Testing requires a specialized lab method; treatment typically involves granular activated carbon or reverse osmosis systems certified specifically for PFAS reduction.

Iron and manganese

Both are naturally occurring, common in groundwater, and primarily aesthetic issues rather than health risks at typical levels: reddish-brown (iron) or black-brown (manganese) staining on fixtures and laundry, metallic taste, and buildup in pipes and water heaters. Manganese at very high levels has been associated with some health considerations, particularly for infant exposure, so persistent elevated manganese is worth discussing with your lab or a water treatment professional rather than dismissing purely as cosmetic.

Hardness

Calcium and magnesium dissolved from rock as groundwater passes through it. Not a health concern, but hard water causes scale buildup in pipes and appliances, reduces soap and detergent effectiveness, and can shorten the lifespan of water heaters and plumbing fixtures over years. Water softeners (Chapter 6) address this directly.

Sulfate and hydrogen sulfide ("rotten egg" smell)

Sulfate can cause a bitter taste and, at high levels, a laxative effect in people unaccustomed to it. Hydrogen sulfide gas produces the classic "rotten egg" smell, often from naturally occurring sulfur bacteria in the well or plumbing system, and while unpleasant, it's usually more of a nuisance than a health hazard, though persistent strong odor warrants testing to rule out other issues and confirm the source.

The bottom line: Bacteria and nitrate are the two contaminants that demand annual, non-negotiable testing because of acute health risk. Arsenic, lead, radon, and PFAS are typically slower-building, region-specific risks worth periodic baseline testing. Iron, manganese, and hardness are common and usually aesthetic rather than dangerous, but still worth addressing for your plumbing and your sanity.

Chapter 5: Choosing a Certified Lab and Collecting a Sample Correctly

A water test is only as good as the sample and the lab behind it. A contaminated sample or an uncertified lab can give you a false sense of security or a false alarm, either of which wastes time and money.

Why certification matters

Use a lab certified by your state's environmental or health department for drinking water analysis. Certification means the lab follows validated methods, participates in proficiency testing, and is accountable for its results. A local hardware store test kit or an uncertified mail-in service may be fine for a rough aesthetic check (hardness, pH) but shouldn't be relied on for health-based parameters like bacteria, nitrate, arsenic, or lead. Your state health department or environmental agency website typically maintains a list of certified labs; many county health departments also offer basic bacteria/nitrate test kits directly, often at a lower cost than private labs.

Collecting a bacteria sample correctly

Bacteria sampling is the most sensitive to collection error, since any bacteria introduced during collection (rather than present in the well) will produce a false positive.

1. Choose a cold-water tap that draws directly from the well line, ideally before any treatment system or softener, and remove any aerator or screen from the faucet.
2. Sanitize the tap opening (many kits include a sanitizing wipe or you can use a flame briefly on a metal tap), then run cold water for several minutes to flush the line.
3. Open the sterile sample bottle only at the moment of filling, without touching the inside of the bottle or cap.
4. Fill to the indicated line without overfilling or rinsing the bottle first (many sample bottles contain a preservative and should not be rinsed).
5. Cap immediately, label with date/time, and refrigerate or keep on ice until delivered to the lab, following the lab's specific holding-time instructions (often within 24-30 hours for bacteria).

Collecting a chemical/metals sample

Chemical parameters are generally more forgiving of minor technique variation than bacteria, but still follow the lab's specific instructions, which often include:

- Running water for a few minutes before sampling (to get a representative sample of the aquifer, not stagnant pipe water) for most chemical parameters
- Using a "first draw" sample instead (water sitting in the pipe overnight) specifically when testing for lead, since the point is to capture worst-case plumbing exposure
- Using the specific bottle and any preservative the lab provides for each parameter; some tests require an acid preservative already in the bottle
- Avoiding contamination from your hands, the tap exterior, or airborne dust during collection

Where to sample from

Sample from a tap before any treatment system whenever you want to know what's actually coming out of the aquifer. Sample from a tap after treatment when you want to confirm a treatment system is working as intended. Testing only after treatment and assuming that reflects your raw well water is a common and easily avoided mistake.

Mail-in vs. local drop-off

Many certified labs offer mail-in kits with pre-paid shipping, convenient for bacteria and standard panels, though holding-time limits matter for bacteria specifically (a sample sitting in transit too long can produce unreliable results). Local drop-off, where available through your county health department or a nearby certified lab, minimizes this risk. If you're mailing a bacteria sample, use expedited shipping and ship early in the week to avoid it sitting in transit or a facility over a weekend.

The bottom line: Use a state-certified lab, follow its specific sample collection instructions exactly (especially timing and technique for bacteria), and know whether you're sampling raw well water or post-treatment water, since testing the wrong point in your system is one of the most common and misleading mistakes well owners make.

Chapter 6: Treatment Systems — Matching the Fix to the Problem

The single biggest treatment mistake well owners make is buying a system before knowing exactly what's in their water. Every treatment technology on this list solves specific problems; none of them solves everything, and installing the wrong one wastes money without fixing the actual issue.

Whole-house vs. point-of-use

- **Whole-house (point-of-entry) systems** treat all water entering the house, appropriate for problems that affect plumbing, appliances, bathing, and drinking alike: hardness, iron, manganese, sediment, and pH correction.
- **Point-of-use systems** treat water at a single tap, typically the kitchen sink, appropriate for problems you only need solved for drinking and cooking water: many reverse osmosis systems, some carbon filters, and some arsenic or nitrate-specific systems.

Matching technology to contaminant

Problem	Common treatment
Bacteria (positive test, ongoing risk)	Shock chlorination (immediate), continuous chlorination or UV disinfection (ongoing)
Nitrate	Reverse osmosis, anion exchange, or distillation; whole-house treatment for nitrate is uncommon, more typically point-of-use for drinking water
Arsenic	Reverse osmosis, adsorptive media (activated alumina, iron-based media), or anion exchange, depending on arsenic form and concentration
Lead/copper (plumbing-sourced)	Corrosion control (pH adjustment), point-of-use filtration certified for lead reduction, and plumbing remediation
Radon	Aeration systems, specifically designed to off-gas radon before water reaches taps
PFAS	Granular activated carbon or reverse osmosis systems specifically certified for PFAS reduction; not all carbon filters are adequate for this
Iron and manganese	Oxidizing filters, water softeners (for lower levels), or specialized iron filters for higher concentrations
Hardness	Ion-exchange water softener
Sediment/turbidity	Sediment pre-filters, often paired with other treatment as a protective first stage
Hydrogen sulfide odor	Aeration, oxidizing filtration, or chlorination followed by filtration, depending on severity
Low pH (corrosive water)	Acid-neutralizing filters (calcite/soda ash) or chemical feed pumps

UV disinfection

Ultraviolet disinfection systems use UV light to inactivate bacteria and other pathogens as water passes through, without adding chemicals. It's a common and effective ongoing solution after a well tests positive for bacteria and the source has been addressed, or as a precaution for wells with a history of contamination risk. UV systems require the water to be relatively clear (high turbidity or iron can shield organisms from the UV light), so they're often paired with a sediment or iron pre-filter, and the UV bulb needs periodic replacement (typically annually) to remain effective, a maintenance step that's easy to forget since the system gives no obvious sign when the bulb has degraded.

Reverse osmosis

A membrane-based filtration system that removes a broad range of dissolved contaminants, including nitrate, arsenic, lead, and many others, typically installed as a point-of-use system under the kitchen sink. It produces highly purified water but also wastes some water in the process and requires periodic membrane and filter replacement. It's one of the most broadly effective treatments available for health-based contaminants, which is why it's a common recommendation when a test reveals multiple issues rather than one isolated problem.

Water softeners

Ion-exchange softeners swap calcium and magnesium ions for sodium (or potassium) ions, eliminating hardness. They require periodic salt (or potassium chloride) refills and occasional resin bed maintenance. Households on a sodium-restricted diet sometimes opt for potassium chloride instead of sodium, or install a reverse osmosis system at the drinking-water tap to avoid the added sodium in softened water.

Getting the sizing and design right

Treatment systems are sized based on your household's water usage, the specific contaminant concentrations in your test results, and your well's flow rate. An undersized system won't keep up with demand; an oversized one wastes money. This is a case where a reputable water treatment professional, working from your actual lab results rather than a generic recommendation, earns their fee. Get a treatment recommendation only after you have test results in hand, never before.

A word on treatment before you have results

Some companies offer a "free water test" as a sales tool, then recommend a system based on an in-home test kit that isn't run through a certified lab. Be skeptical of any treatment recommendation that isn't backed by results from a certified laboratory, and be especially skeptical of a company that both tests your water and sells you the system, without an independent lab result to confirm the problem first. This isn't automatically a scam, some legitimate companies do both, but it's a setup where getting a second, independent lab test before committing to a system is cheap insurance.

The bottom line: Treatment should always follow a certified lab test, never precede it. Match the technology specifically to the contaminant your results show, understand whether you need whole-house or point-of-use

treatment for that specific problem, and get an independent lab-confirmed baseline before trusting any company's in-home test as the basis for a purchase.

Chapter 7: Troubleshooting Taste, Smell, Color, and Pressure Problems

This chapter walks through common symptoms, likely causes, and reasonable first steps, along with when to test and when to call a professional.

Problem: Metallic or "rusty" taste, reddish-brown staining

Step 1. This pattern usually points to iron. Check whether staining appears mainly on white fixtures, laundry, and around faucet aerators.

Step 2. Test for iron and manganese to confirm the concentration.

Step 3. If confirmed, an oxidizing filter or iron-specific treatment system is the typical fix (Chapter 6). This is generally an aesthetic issue, not urgent, but worth addressing before it damages fixtures and appliances further.

Problem: "Rotten egg" smell

Step 1. Note whether the smell is present in hot water only, cold water only, or both. Hot-water-only odor often points to a water heater anode rod reacting with sulfate, a plumbing issue rather than a well issue.

Step 2. If present in both hot and cold, it's more likely hydrogen sulfide from the well or plumbing system generally.

Step 3. Test if the odor is new, worsening, or hasn't been evaluated before. Treatment options are covered in Chapter 6.

Problem: Sudden taste, smell, or color change

Step 1. Treat this as a reason for an immediate, out-of-cycle bacteria and general test, not just an annoyance. A sudden change often means something entered the well or aquifer that wasn't there before.

Step 2. Avoid drinking the water until you have a test result, especially if you notice the change alongside cloudiness, an unusual odor, or any household illness.

Step 3. If the change followed a specific event (nearby flooding, well work, a chemical spill in the area), mention that specifically to your lab, since it can help target the right test panel.

Problem: Cloudy or discolored water

Step 1. Check whether it clears after running the tap for a minute or two (often trapped air, generally harmless) versus persistent cloudiness or color.

Step 2. Persistent cloudiness can indicate sediment, a failing well component, or in some cases a sign the well has been disturbed by nearby construction or a dropping water table. Test and, if it persists, have the well itself inspected by a licensed well contractor.

Problem: Low water pressure or the well "running dry"

Step 1. Check whether other household plumbing issues (a partially closed valve, a failing pressure tank) could explain it before assuming a well problem.

Step 2. A well that runs dry, especially seasonally during dry periods, may indicate the well doesn't have sufficient yield for the household's usage, or the water table has dropped. This is a well-contractor issue (well depth, pump placement) rather than a water-quality issue, though it's worth mentioning if you're also troubleshooting quality changes, since a low water table can concentrate certain contaminants.

Step 3. Persistent or worsening low pressure warrants a call to a licensed well or pump contractor.

Problem: Positive bacteria test

Step 1. Stop drinking the water without boiling it first (a rolling boil for at least one minute, longer at high altitude) or switch to bottled water, especially for infants, pregnant women, or anyone immunocompromised.

Step 2. Inspect the well cap and casing for obvious damage or gaps, and check for any recent nearby flooding or construction.

Step 3. Shock chlorinate the well (many state health departments publish a step-by-step protocol, or hire a licensed well contractor to do it) and retest after the recommended waiting period, typically one to two weeks after flushing the chlorine out.

Step 4. If retesting still shows bacteria present, or if it recurs repeatedly, get a licensed well contractor to inspect the well itself for a structural pathway (cracked casing, failed cap seal, improper grouting) rather than repeating shock chlorination indefinitely as a band-aid.

Quick reference table

Symptom	Likely cause	Action
Rusty taste/staining	Iron	Test, then treat with oxidizing filter
Rotten egg smell, both hot/cold	Hydrogen sulfide	Test, treat per Chapter 6
Rotten egg smell, hot only	Water heater anode rod	Plumbing fix, not a well issue
Sudden taste/smell/color change	Possible new contamination	Test immediately, avoid drinking until confirmed safe
Persistent cloudiness	Sediment or well disturbance	Test, consider well inspection
Low pressure / dry well	Yield or pump issue	Call a licensed well/pump contractor
Positive bacteria test	Well cap, casing, or nearby contamination	Boil water, shock chlorinate, retest

The bottom line: Most taste, smell, and color changes have identifiable causes and reasonable first steps, but any sudden change or a positive bacteria result should be treated as a reason to stop drinking untested water immediately, not something to wait out. When in doubt, test first and decide second.

Chapter 8: Seasonal Care and Flood/Emergency Protocols

Well water quality and well function both interact with weather and season more than most owners expect.

Spring and post-snowmelt

Spring snowmelt and rain can temporarily raise the water table and increase the risk of surface contaminants reaching a shallow well or entering through a compromised cap or casing. It's a reasonable time to do a visual check of your well cap and casing seal each year, and if your well has a history of any bacteria concerns, consider it a good annual testing window.

Flooding

If your well has been flooded or the area immediately around it was underwater:

- **Do not use the water for drinking, cooking, or brushing teeth** until it's been tested and, if needed, disinfected.
- Once floodwaters recede and the area around the well is no longer submerged, have the well professionally inspected before resuming normal use, particularly if the well cap or any electrical components were underwater.
- Shock chlorination is typically required after a flood event, followed by testing to confirm the water is safe before returning to normal use.
- If the pump or electrical system was submerged, have it inspected by a licensed electrician or well contractor before running it; submerged pump motors and controls can pose both a function risk and a shock hazard.

Drought and low water table periods

- Reduced aquifer recharge during drought can concentrate naturally occurring minerals and contaminants, since there's less fresh water diluting the aquifer.
- A well running low or "sputtering" during drought may indicate the water table has dropped closer to the pump intake, worth having a well contractor assess before it becomes a complete loss of supply.
- It's a reasonable time to be more conservative with water use generally, spacing out heavy-use activities like laundry and lawn irrigation.

Winter and freeze protection

- Well pumps themselves are typically buried below the frost line or housed in a heated well house/basement and generally aren't at high freeze risk, but exposed pipes running from the well to the house, especially in unheated crawlspaces, are vulnerable.
- Insulate exposed pipes and consider heat tape for particularly vulnerable runs in cold climates.

- If a property will sit unoccupied through a hard freeze, have a professional advise on winterizing the well and plumbing system specifically, not just the house's interior plumbing.

Emergency situations: what to do if you suspect contamination

1. Stop using the water for drinking, cooking, and brushing teeth immediately.
2. Boil water if you must use tap water for these purposes, or switch entirely to bottled water.
3. Contact your state or local health department; many maintain guidance and emergency testing resources specifically for private well contamination events.
4. Get a certified lab test as soon as possible to confirm the specific issue.
5. Don't resume normal use until you have a clean test result following any treatment (shock chlorination, filter installation, etc.) that was needed.

Seasonal and vacation properties

A property used only part of the year presents specific well considerations: a well sitting unused for months can develop stagnant water in the pipes and pressure tank, worth flushing thoroughly (run taps for several minutes) before resuming normal use after a long vacancy, and worth testing if the vacancy was extended (several months or more) before trusting the water again without question.

The bottom line: Flooding and drought both create elevated risk windows for well water quality, flooding through direct contamination and drought through concentration of existing minerals and lower water tables. Never use well water for drinking or cooking after flooding without testing and likely disinfection first, and treat a long vacancy as a reason to flush and consider retesting before resuming normal use.

Chapter 9: Buying or Selling a Home with a Private Well

A private well changes the due-diligence process on both sides of a real estate transaction, and it's an area where a modest testing cost upfront can prevent a significant health or financial surprise later.

If you're buying

Get a dedicated well water test, separate from the general home inspection. Many general home inspectors don't test water quality at all, or only check flow rate and basic function. A dedicated water test, at minimum covering coliform bacteria, nitrate, and a broader panel appropriate to your region (arsenic, PFAS, radon where regionally relevant), gives you the picture a general inspection won't.

Ask for testing history. A seller with regular test records is a good sign. No records at all isn't automatically a red flag, wells change hands with incomplete paperwork often, but it means you're buying more uncertainty and should weight your own testing accordingly.

Ask about the well's age, depth, and type. An older well, a shallow dug well, or one without a documented, sealed casing carries different risk than a newer, deep, properly cased drilled well. This should inform how thorough your testing is and what you budget for potential treatment.

Understand the well's yield. Ask whether the well has ever run dry or shown signs of low yield, particularly during summer or drought periods, since this affects whether it can support your household's actual water use, not just the previous owner's.

Check well and septic setback distances if the property also has a septic system. Wells and septic systems too close together (distances vary by state and local code) carry elevated contamination risk regardless of how either individually tests today.

Negotiate based on findings. A bacteria-positive result is often resolvable with shock chlorination before closing, a relatively minor item. Elevated arsenic, PFAS, or a well with confirmed low yield are bigger conversations, and you have real leverage to negotiate treatment system installation, a price adjustment, or in some cases walk away.

If you're selling

Test before listing. A recent, clean test result with paperwork in hand heads off buyer concerns early and can be a genuine selling point.

Disclose what you know. Disclosure requirements vary by state, but known issues (a past bacteria event, elevated minerals, any treatment system installed to address a specific problem) are better disclosed upfront than discovered by a buyer's inspector.

Gather your records. Pull together your testing log, well construction record or permit if you have it, and any treatment system documentation (install date, maintenance history, filter/media replacement schedule).

Have any treatment systems serviced and documented before showings, since a well-maintained treatment system with clear service records reads very differently to a buyer than an unexplained system with no history.

A note on required testing

Many states and counties require well water testing as part of a real estate transaction, sometimes for specific parameters only (often coliform and nitrate at minimum), sometimes more broadly depending on local rules. Check your state and county requirements early in the process, since a required test that turns up a problem can affect your closing timeline.

The bottom line: A dedicated well water test, separate from the general home inspection, is essential on both sides of a sale. Buyers should treat missing test records and unknown well age/type as reasons for more thorough testing, not less, and sellers who test proactively and disclose known issues generally have smoother, faster transactions.

Chapter 10: Costs & Budgeting

Well water testing and treatment costs range widely, and the biggest factor in your long-term expense is whether you catch and address problems early through regular testing rather than discovering them as a crisis.

Typical testing cost ranges (2026 estimates)

Test	Typical range
Basic bacteria + nitrate package	\$30–\$60
Broader panel (hardness, iron, manganese, pH, TDS)	\$60–\$150
Arsenic (standalone)	\$20–\$50
Radon in water	\$50–\$100
PFAS panel	\$250–\$600
Lead and copper	\$30–\$75
Comprehensive/full panel (dozens of parameters)	\$150–\$400

Typical treatment cost ranges (2026 estimates)

System	Typical range (installed)
Sediment pre-filter	\$150–\$500
Whole-house carbon filter	\$500–\$2,000
Water softener	\$800–\$2,500
Iron/manganese filtration system	\$1,000–\$3,500
UV disinfection system	\$500–\$1,500
Reverse osmosis (point-of-use, under-sink)	\$200–\$800
Whole-house reverse osmosis	\$3,000–\$10,000+
Radon aeration system	\$2,000–\$5,000
PFAS-certified treatment system	\$1,500–\$6,000+
Shock chlorination service	\$150–\$400
Well cap/casing repair	\$200–\$1,500
New well (full drilling, varies enormously by depth/geology)	\$5,000–\$25,000+

These figures reflect general US ranges for 2026 and move significantly with your region, well depth, contaminant concentration, and site accessibility. Always get local quotes, particularly for anything beyond routine testing.

The repair vs. treat vs. re-drill decision framework

When a test reveals a problem, a useful way to frame the decision:

- 1. Is this a plumbing issue or a source-water issue?** Lead often comes from your plumbing, not the aquifer. Iron and arsenic are typically source water. This changes whether you're fixing pipes or treating incoming water.
- 2. Is the concentration close to the limit or far above it?** A nitrate result at 11 mg/L against a 10 mg/L limit is a different urgency level than one at 40 mg/L, even though both technically exceed the standard.
- 3. Is this a one-time event or an ongoing condition?** A bacteria positive after known flooding, resolved with shock chlorination and a clean retest, is different from recurring bacteria detections that point to a structural well problem needing real repair.
- 4. What does the cost comparison actually look like?** Compare treatment system cost against realistic alternatives: bottled water indefinitely, a new well, or connecting to municipal water if it's available nearby. Treatment is usually, but not always, the most cost-effective long-term answer.

Budgeting for the long term

A practical approach: budget for annual testing as a fixed, small cost (treat it like an insurance premium), and set aside a modest reserve for eventual treatment system installation or maintenance, since most private well owners will need at least one treatment system at some point given how common iron, hardness, and regional contaminants are. Pair that with the maintenance habits in this book, and the goal isn't zero cost, it's staying ahead of problems instead of discovering them during a health scare or a home sale.

The bottom line: Routine annual testing costs less than a nice dinner out; treating a confirmed problem costs hundreds to a few thousand dollars in most cases; a truly neglected well (recurring contamination, failed casing, insufficient yield) can cost tens of thousands to fully resolve. Testing regularly and treating specific, confirmed problems keeps you in the cheap tier.

Chapter 11: Records, DIY vs. Pro, and Building a Relationship with a Water Professional

Why records matter more than people think

A well's water quality history is largely invisible without documentation. You can't see the aquifer, you can't see what's happening underground upstream of your well, and most owners don't remember exactly when they last tested without checking a note somewhere. Good records solve this. They let you spot trends (rising nitrate over several years, for instance) before they cross a threshold, they support your case in a real estate transaction (Chapter 9), and they give any water treatment professional a real starting point instead of a guess.

At minimum, keep: every test result with date and lab, the well construction record or permit if you have it (often available from your state or county even if you don't have a personal copy), treatment system installation and maintenance records, and notes on any well work (new pump, casing repair) done over the years.

DIY vs. professional: where the line sits

Reasonable for a homeowner to handle: - Visual inspection of the well cap and casing for damage or poor sealing - Collecting a water sample correctly, following your lab's instructions - Keeping the testing log and treatment maintenance schedule - Replacing certain filter cartridges and treatment system consumables per the manufacturer's instructions (many systems are designed for this) - Basic drainage management around the wellhead (keeping it sloped away from the casing)

Always call a licensed well or water treatment professional for: - Interpreting an unusual or concerning lab result and recommending treatment - Well cap, casing, or structural repair - Pump installation, repair, or replacement - Shock chlorination, especially after a positive bacteria result or flooding (many state health departments provide DIY protocols, but professional service is often worth it for reliability, especially after a repeat positive) - Installing whole-house treatment systems, especially anything involving electrical or plumbing modification - Diagnosing yield or pressure problems that could indicate a failing pump or dropping water table

Warranties

If you've had a treatment system installed, ask specifically what's under warranty, for how long, and what maintenance is required to keep the warranty valid (many systems require documented filter or media changes on schedule). Get warranty terms in writing and note the expiration and required maintenance intervals in your log.

Building a relationship with a water professional

There's real value in using the same certified lab and, where you have one, the same water treatment company over years. A company familiar with your well's history can spot a meaningful change faster than one seeing your results for the first time, and can advise you on whether a shift is a normal seasonal fluctuation or a genuine new problem. That said, get a second opinion before committing to a major treatment system purchase, particularly if a single

company both tested your water and is recommending the system, since an independent confirmation is cheap insurance against an unnecessary sale.

Putting it together

The well owners who spend the least over the years, and worry the least about their water, aren't the ones with the fanciest treatment system. They're the ones who test every year without fail, understand what their results mean well enough to spot a real change, treat the specific problems their water actually has, and keep records that turn a decade of scattered results into a clear picture. None of that requires expertise. It just requires a yearly habit.

The bottom line: Keep a complete testing and treatment record, know which tasks are safe to handle yourself and which need a licensed professional, and get an independent second opinion before committing to a major treatment purchase. A consistent testing habit, more than any single piece of equipment, is what keeps a private well safe over a lifetime of ownership.

Appendix A: Printable Testing Log

Copy this table into your own document, print it, or keep it digitally. Fill in a new row every time you test.

Date	Lab used	Bacteria result	Nitrate (mg/L)	Other parameters tested	Results/notes	Cost

System reference information (fill in once):

Field	Your well's info
Well type (drilled/dug/driven point)	
Well depth (if known)	
Well age / drilling year	
Casing material	
Number of people in household	
Treatment systems installed (list all)	
Location of well cap / access	
Well permit / construction record location	

Appendix B: Glossary of Terms

Aquifer: An underground layer of saturated rock, sand, or gravel that stores and transmits groundwater; the source your well draws from.

Coliform bacteria: A group of bacteria used as an indicator of possible contamination in water; presence suggests a pathway exists for surface or fecal contamination to reach the well.

MCL (Maximum Contaminant Level): The legally enforceable limit the EPA sets for a given contaminant in public water systems; not legally binding for private wells but used as a health-based reference point.

Non-detect (ND): A lab result indicating a substance wasn't found above the test method's detection limit.

PFAS: Per- and polyfluoroalkyl substances, a family of persistent synthetic chemicals sometimes called "forever chemicals," found in some firefighting foam, industrial, and consumer product applications.

Point-of-entry (whole-house) treatment: A treatment system installed where water enters the house, treating all water used in the home.

Point-of-use treatment: A treatment system installed at a single tap or fixture, treating only water drawn from that point.

Primary standard: An EPA health-based, legally enforceable standard for public water systems.

Secondary standard: An EPA aesthetic (non-health-based, non-enforceable) guideline covering taste, odor, color, and similar quality factors.

Shock chlorination: A disinfection process introducing a strong chlorine solution into a well to kill bacteria, typically followed by flushing and retesting.

Turbidity: The cloudiness of water caused by suspended particles; can shield bacteria from UV disinfection if high enough.

Water table: The upper surface of the saturated zone in an aquifer; its depth can rise or fall seasonally and during drought.

Appendix C: 20 Questions to Ask Before You Hire a Water Treatment Company

1. Do you require a certified lab test before recommending a treatment system, or do you rely only on your own in-home test?
 2. Can I see the certified lab results you're basing your recommendation on?
 3. Are you licensed or certified to install water treatment systems in this state?
 4. Can you explain specifically which contaminant each recommended system addresses?
 5. Is this whole-house or point-of-use treatment, and why is that the right choice for my results?
 6. What size/capacity system do you recommend for my household, and how did you calculate that?
 7. What's the total installed cost, and what does that include?
 8. What ongoing maintenance does this system require, and how often?
 9. What's the annual cost of consumables (filters, media, salt, UV bulbs)?
 10. What warranty do you offer, and what maintenance is required to keep it valid?
 11. Can you provide references from customers with a similar well and water profile to mine?
 12. Will you provide written documentation of the installation and system specifications?
 13. Do you recommend retesting after installation to confirm the system is working, and will you help arrange that?
 14. If my results change over time, how do I know whether the system needs adjustment?
 15. Is this system independently certified (NSF/ANSI standards) for the specific contaminant I'm treating?
 16. What happens if the system fails, in terms of response time and cost?
 17. Do you offer financing, and what are the terms?
 18. Are you the same company that performed my water test, and if so, can you point me to an independent lab confirmation as well?
 19. How long have you been installing systems in this specific region, and are you familiar with our local geology/water issues?
 20. What would you recommend if the primary issue I'm treating for turns out not to be the biggest problem in my results?
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Closing Note

Well ownership isn't difficult once you know the handful of things that matter: test every year without exception for bacteria and nitrate, understand what your results actually mean before assuming the worst or dismissing a real problem, and treat the specific issue your water has rather than whatever a salesperson is pushing. Keep this book somewhere you'll actually find it again, whether that's a shelf, a saved file, or a printed copy next to your test kit.

For free updates, seasonal reminders, and further guides, visit **WellWater HQLabs.com/guides**.

This book provides general educational guidance and is not a substitute for a certified laboratory water test, professional water treatment advice, or medical guidance specific to your household's health.